computations in a single step, 'n' being the number of qubits available.

If a particle interacts with another particle at some time, it retains some connection with the other particle, and they can be later correlated with each other in a process called entanglement. In this process, if one particle is in a spin-up position, then the correlated or entangled particle can be forced to be in the opposite state, i.e., spin-down. The spin-state of a qubit under observation is determined at measurement time and is forwarded to the entangled qubit, which at the same time accepts the contrary spin-state to that of the observed qubit. The mechanism of this real phenomenon is not explainable by any theory as yet. Even Einstein called it a 'spooky action at a distance'.

Quantum computing utilises the capability of sub-atomic particles for multi-state existence at any given time. This behaviour of the smallest particles is responsible for energy and time-efficient operations unlike classical computers.

Figure 2: The first fully integrated quantum computer by IBM: IBM Q System One

Figure 3: IBM Q System One flaunting its state-of-art internal structure at CES 2019 in Las Vegas, USA

Quantum vs classical computing

The most evident and striking difference between classical and quantum computing is their fundamental units. The working mechanism of classical bits and qubits makes the two approaches discernibly different from each other. The traditional bit can store either 0 or 1 at a time while a qubit can be both at any time. The total measurement of a qubit at any time is the sum of magnitudes of the spin-up state and the spin-down state. Table 1 highlights a few important points that clearly differentiate classical and quantum computing.

Research scope in quantum computing

Quantum computing is an upcoming yet challenging research area. With the need to build quantum computers with more and more qubits to take the research forward, we greatly rely on technology giants like IBM and Google. The availability of a quantum computer was a dream until IBM built its own models — IBM Q (Figures 2 and 3) and allowed the world to access them remotely on a cloud server. From simulations to actual quantum processing, IBM Q provides a platform for budding researchers to experiment their way into this magical field. With the promises and challenges of quantum computing, there's a lot that is yet to be discovered.

- **What's been done?**
  Physicists and researchers across the world have discovered many quantum algorithms in the 1980s and 1990s that gained a lot of appreciation. Still, in the absence of an actual quantum computer, it remained a dream to actually implement any of these. Peter Shor, in 1994, took the world by surprise with his quantum algorithm for integer factoring. Despite theoretical progress in the field in the 1990s, scientists still believed that an entire fault-tolerant quantum computing was a distant dream.

- **What's being done?**
  On October 2019, Google AI, in partnership with NASA, claimed it had achieved 'quantum supremacy'. This term, originally coined by John Preskill in 2012, describes a point where quantum computers can carry out tasks that classical computers cannot (within a feasible amount